

Notes: Bybee, Kelly & Su (RFS, 2023)
Narrative Asset Pricing: Interpretable Systematic Risk Factors from News Text

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1 Big picture

- **Question.** What are the fundamental risks priced in the cross-section of stock returns, and can business-news narratives reveal interpretable ICAPM state variables?
- **Core premise.** The ICAPM says risk premia compensate exposures to shocks to wealth and future investment opportunities, but these state variables are hard to observe.
- **Data innovation.** The paper uses Wall Street Journal article text to construct narrative attention shocks.
- **Method.** The model integrates LDA topic modeling, IPCA latent factor analysis, and group-lasso variable selection in a Sparse IPCA framework.
- **Result.** Narrative factors have lower pricing errors and higher out-of-sample MVE Sharpe ratios than standard benchmark factors.
- **Interpretability.** The model maps factors back to narratives, words, and articles; Recession has the strongest negative impact on the pricing kernel, while Record High and Optimism are positive.

2 Incremental contribution

- Introduces WSJ business narratives as empirical proxies for latent ICAPM state variables.
- Embeds text directly in an asset pricing model rather than using text only for prediction.
- Develops Sparse IPCA, combining IPCA with group lasso to select economically relevant narratives.
- Provides a factor interpretation pipeline from latent factors to narratives, words, and articles.
- Shows that narrative factors add value relative to Fama-French-Carhart factors and forecast macro variables in ICAPM-consistent directions.

3 Data and measurement

The text source is Wall Street Journal article text from 1984:01 to 2017:06. The authors use business-oriented sections and filter out nonbusiness content. LDA estimates 180 narrative topics. Article-level topic attention θ_m is averaged into daily attention θ_τ , and attention shocks z_τ are deviations from a trailing moving average. Stock returns come from CRSP from 1984:01 to 2016:12. The main monthly sample spans 1985:01–2016:12 and contains 1,850,401 stock-month observations for 15,831 stocks. Test assets include 78 anomaly portfolios and 25 size/book-to-market portfolios. Benchmark factors are CAPM, FF3, FF5, and FFC6.

4 Models and empirical specifications

4.1 News side

$$z_\tau = Ax_\tau + \eta_\tau.$$

Detailed interpretation: The matrix A maps latent ICAPM state variables into narrative attention shocks. Irrelevant narratives have rows of A near zero.

4.2 LDA text model

$$w_m \sim \text{Mult}(\Phi\theta_m, N_m).$$

Detailed interpretation: LDA converts the high-dimensional word-count vector into topic attention weights and topic-term probabilities.

4.3 Return side

$$r_{i,t+1} = \beta'_{i,t} f_{t+1} + \varepsilon_{i,t+1}.$$

Detailed interpretation: f_t is the return-space projection of ICAPM state variables and is used as the systematic risk factor.

4.4 Narrative covariance instruments

$$\text{cov}_{i,t} = \text{Cov}_t(r_{i,t+1}, z_{t+1}), \quad \beta_{i,t} = \text{cov}_{i,t} \Gamma.$$

Detailed interpretation: A stock's covariance with narrative shocks identifies its exposure to systematic narrative risk.

4.5 Sparse IPCA objective

$$\min_{\Gamma, \{f_t\}} \frac{1}{2} \sum_{i,t \in S} (r_{i,t} - c_{i,t-1} \Gamma f_t)^2 + \lambda N_S \sum_{l=0}^L \sigma_l^c \|\Gamma_l\|_2 + \sum_{t \in S} \|f_t\|_2^2.$$

Detailed interpretation: The group-lasso penalty selects narratives by shrinking entire rows of Γ to zero.

4.6 Macro predictive regression

$$\sum_{s=1}^h \psi_{t+s} = b_h (x_t^{MVE} / \text{std}(x_t^{MVE})) + \text{error}_t^{(h)}.$$

Detailed interpretation: The estimated pricing kernel should forecast future investment opportunities in directions consistent with the ICAPM.

5 Identification and empirical design

The identifying content comes from cross-sectional differences in stock covariances with narrative shocks and time variation in narrative attention. Sparse IPCA filters the 180 candidate topics into a smaller set of priced narratives. Out-of-sample tests reduce concern that the factors merely overfit returns. Macro predictive regressions check whether the estimated state variable behaves like an ICAPM pricing kernel. The design does not prove that WSJ news causally moves returns; it shows that news narratives can be used to construct interpretable priced risk factors.

6 Section-by-section study guide

- **Introduction.** Motivates the use of business news to measure latent ICAPM state variables.
- **Model.** Links latent state variables, narrative shocks, stock returns, and article word counts.

- **Estimation.** Introduces Sparse IPCA and Sharpe-ratio tuning.
- **Empirical results.** Evaluates pricing errors, MVE Sharpe ratios, and benchmark comparisons.
- **ICAPM validation.** Tests macro forecasting implications.
- **Interpretation.** Maps latent factors to narratives, terms, and articles.

7 Tables and figures

7.1 Figure 1: Illustration of the data-generating process

Read: The diagram links latent ICAPM state variables to narrative attention shocks, return-mimicking portfolios, and LDA word frequencies.

Detailed interpretation: The figure is the blueprint for the paper: the model recovers unobserved state variables from both text and returns.

Economic message: Business news becomes a measurement system for latent asset-pricing state variables.

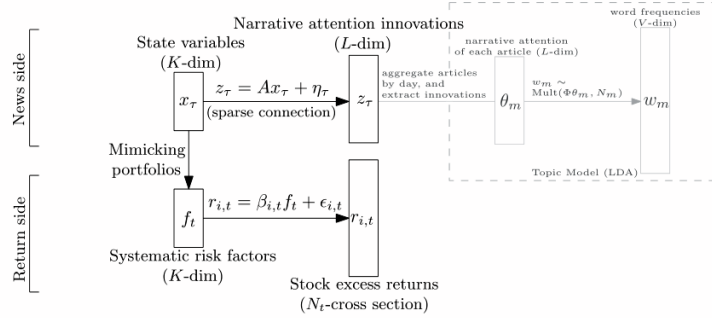


Figure 1
Illustration of the data-generating process
 τ indexes days, t for months, i for stocks, and m for articles. L is the number of narratives, and V is the size of the vocabulary.

Figure 1: Illustration of the data-generating process

7.2 Figure 2: Full-sample estimates with varying λ values + Table 1: Cross-sectional pricing results

Read: Figure 2 shows that modest regularization improves MVE Sharpe ratios while reducing selected narratives. Table 1 shows smaller pricing errors for narrative factors relative to standard benchmarks.

Detailed interpretation: Sparse IPCA improves economic factor quality by reducing overfit and retaining narratives with pricing value.

Economic message: Narrative factors are useful for both statistical fit and asset-pricing performance.

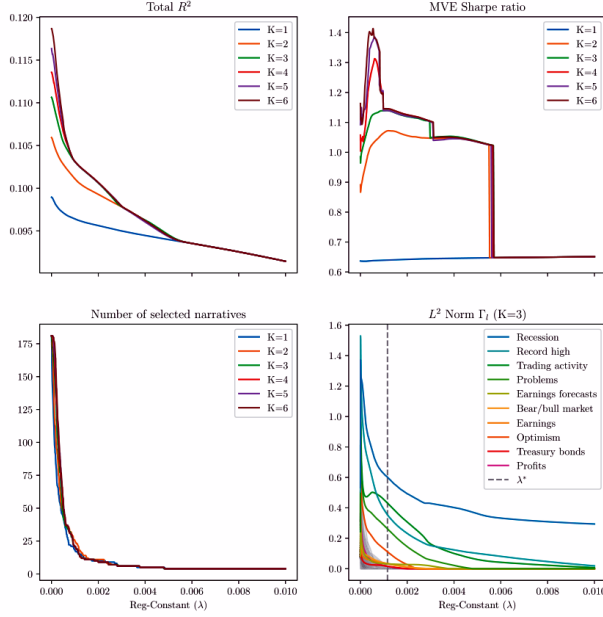


Figure 2

(a) Figure 2

Table 1
Cross-sectional pricing results

A. 78 anomaly portfolios as test assets					B. 25 sizebm double sorts as test assets				
Factors	avg $ \hat{\alpha}_a $	avg $ t(\hat{\alpha}_a) $	$\frac{\# t(\hat{\alpha}_a) > 1.96}{\# \text{ test assets}}$	GRS	Factors	avg $ \hat{\alpha}_a $	avg $ t(\hat{\alpha}_a) $	$\frac{\# t(\hat{\alpha}_a) > 1.96}{\# \text{ test assets}}$	GRS
Mkt	1.11	2.64	0.54	8.56	Mkt	0.42	3.03	0.84	11.61
FF3	0.97	2.65	0.54	8.50	FF3	0.32	3.89	0.84	15.12
FF5	1.18	3.20	0.68	7.48	FF5	0.27	3.31	0.72	13.06
FFC6	1.27	3.43	0.74	7.41	FFC6	0.28	3.40	0.76	12.68
NF1	1.29	3.03	0.73	8.78	NF1	0.35	2.33	0.64	6.56
NF2	0.97	2.72	0.54	7.98	NF2	0.16	1.14	0.16	5.29
NF3	0.84	2.62	0.54	7.84	NF3	0.25	1.95	0.56	5.72
NF4	0.92	2.81	0.55	7.53	NF4	0.23	1.75	0.44	5.58
NF5	0.91	2.78	0.60	7.43	NF5	0.23	1.78	0.48	5.48
NF6	0.96	2.89	0.63	7.38	NF6	0.25	1.91	0.48	5.56

Statistics: (1) avg $|\hat{\alpha}_a|$, cross-sectional average absolute value of the intercepts (a indexes anomaly test assets); (2) avg $|t(\hat{\alpha}_a)|$, the cross-sectional average absolute value of the intercept t -statistics; (3) $\frac{\#|t(\hat{\alpha}_a)| > 1.96}{\# \text{ test assets}}$, the proportion of intercepts statistically significantly different from zero; and (4) GRS, the GRS statistic for the joint test of all intercepts are zero.

(b) Table 1

7.3 Figure 3: L^2 norm of Γ in expanding training samples + Table 2: Out-of-sample Sharpe ratios of MVE portfolios

Read: Figure 3 shows stable narrative importance across training samples. Table 2 shows narrative MVE Sharpe ratios near 1.3, above benchmark factors.

Detailed interpretation: Stability plus out-of-sample performance support the factor interpretation.

Economic message: Narrative factors generalize beyond the estimation sample.

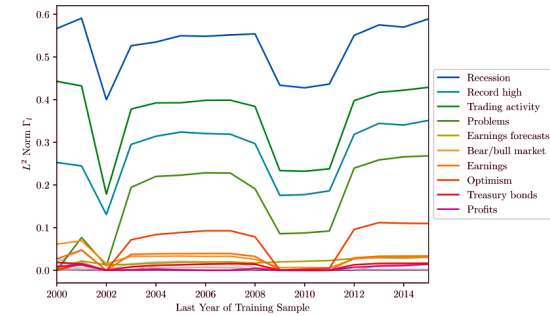


Figure 3

(a) Figure 3

Table 2
Out-of-sample Sharpe ratios of MVE portfolios

A. Narrative factor model		K					
Tuning	Statistics	1	2	3	4	5	6
$\lambda = \lambda_{\frac{1}{25}}^*$	Sharpe ratio	0.48	1.00	1.10	1.26	1.32	1.31
	# narratives	2.9	4.9	12.1	39.1	43.4	61.8
$\lambda = 0$	Sharpe ratio	0.44	0.66	0.73	0.73	0.79	0.91
	# narratives	All 180 narratives, no selection					
B. Benchmark factors		Mkt	SMB	HML	RMW	CMA	UMD
Sharpe ratio		0.25	0.13	0.36	0.82	0.90	0.67

Panel A reports out-of-sample MVE portfolio Sharpe ratios for narrative factor models with up to six factors. The rows $\lambda = \lambda_{\frac{1}{25}}^*$ correspond to Sparse IPCA and the "# narratives" reports the number of selected narratives (averaged over all training samples). The rows $\lambda = 0$ correspond to unregularized IPCA and thus all narratives are included without penalization. In panel B, each column reports the out-of-sample MVE portfolio Sharpe ratio using all factors up to that column (e.g., the column labeled "RMW" reports the MVE combination of Mkt, SMB, HML, RMW).

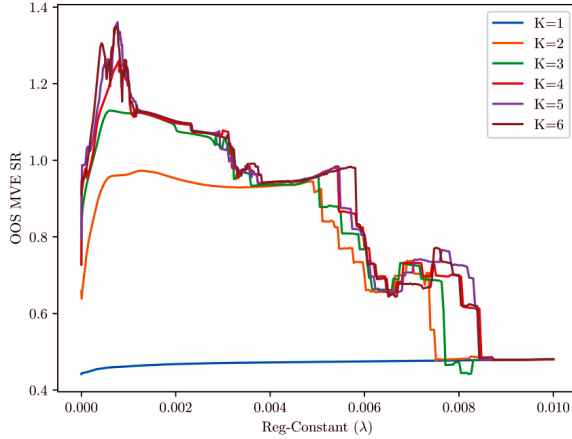
(b) Table 2

7.4 Figure 4: Out-of-sample MVE Sharpe ratio versus λ + Figure 5: Out-of-sample MVE cumulative returns

Read: Figure 4 shows the out-of-sample benefit of regularization. Figure 5 shows cumulative MVE returns for regularized, unregularized, and benchmark portfolios.

Detailed interpretation: The gain from Sparse IPCA is visible in portfolio performance, not just model fit.

Economic message: Regularization improves realized investment value.



(a) Figure 4

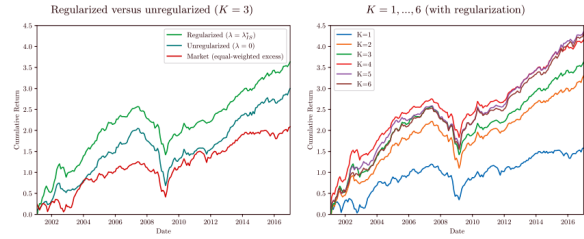


Figure 5
Out-of-sample MVE cumulative returns
Cumulative returns of the OOS f^{MVE} in different specifications. The left panel fixes $K=3$ and plots the benchmark specification in green, the unregularized in blue, and the market factor in red for comparison. The

(b) Figure 5

7.5 Table 3: Sharpe ratios combining FFC6 and NF + Figure 6: Pricing kernel forecasts macro variables

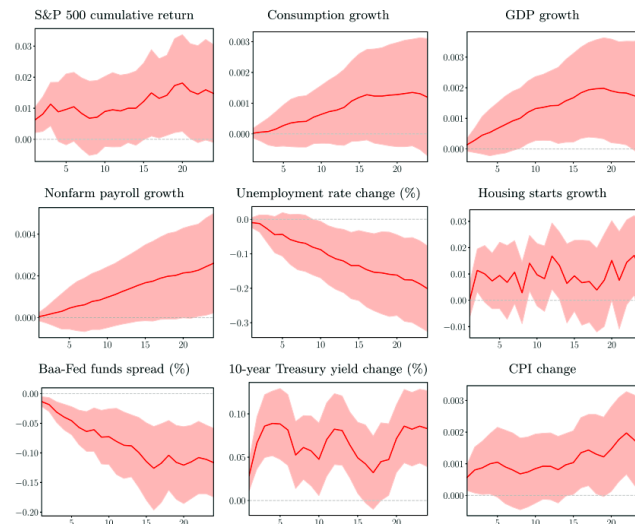
Read: Table 3 shows narrative factors add to FFC6. Figure 6 shows the narrative pricing kernel predicts macro variables with ICAPM-consistent signs.

Detailed interpretation: Narrative factors are not just re-labeled benchmark factors; they contain incremental macro-risk information.

Economic message: The model connects stock returns, news narratives, and future investment opportunities.

Specification	K (number of NF added)						
	0	1	2	3	4	5	6
NF + Mkt		0.48	1.00	1.10	1.26	1.32	1.3
NF + FF3	0.25	0.48	1.00	1.08	1.26	1.32	1.3
NF + FF5	0.36	0.41	0.60	0.68	1.02	1.17	0.90
NF + FFC6	0.90	0.89	0.94	0.92	1.01	1.01	1.00
	0.67	0.65	0.76	0.80	0.87	0.92	1.10

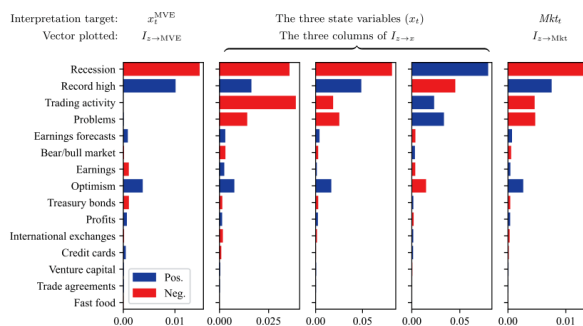
(a) Table 3



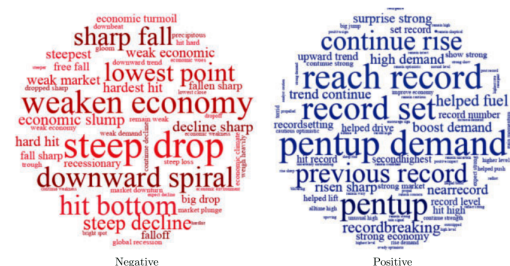
(b) Figure 6

Read: Recession has the largest negative impact on the MVE state variable; Record High and Optimism have leading positive impacts. Term clouds show recessionary and expansionary language.

Economic message: The pricing kernel can be read as a narrative object.



(a) Figure 7



(b) Figure 8

7.7 Table 4: Example news articles + Figure 9: Event retrieval for market returns

Read: Table 4 lists article headlines for Recession, Record High, and Trading Activity. Figure 9 retrieves articles associated with extreme market return shocks.

Detailed interpretation: The model translates statistical risk factors into concrete article-level narratives.

Economic message: Text-based asset pricing can explain both portfolio risk and specific market-moving news events.

Table 4
Example news articles of the three most prominent narratives

Recession	
1993-05-07	Auto Registrations Continued to Slump In Europe Last Month
2001-04-25	Consumer Confidence Slides on Fears of Layoffs
2009-02-19	U.S. News: Housing Starts Hit Lowest Level In Half-Century
2011-08-02	World News: Manufacturing Slowdown Adds to Gloom on Economy
2016-07-08	World News: U.K. Consumer Sentiment Takes Dive
Record high	
1989-07-05	Japan Vehicle Sales Rise
1994-07-01	Purchasing Managers In U.K. Survey Report Rise for June Orders
1995-02-27	Hiring Outlook For Second Quarter Appears Vigorous
2006-01-12	Wall Street Bonuses Hit a Record in 2005
2016-07-20	U.S. News: Home Building Continues Recovery as Demand Rises
Trading activity	
1993-12-30	Industrials Rise A Bit to Record; Bonds Decline
1994-10-20	Profit News Helps Boost Stock Prices — Indexes Gain Ground Despite Weakness Of Bonds and Dollar
1996-06-21	Nasdaq Sinks Amid Sell-Off Of Tech Stocks
1997-12-09	Blue Chips Fall As Dollar's Rise Causes Concern
1998-04-21	Drug Stocks Resume Gains; Blue Chips Fall

This table reports example articles with the highest attention to the three narratives, respectively (titles only; see text bodies in [Internet Appendix D.1](#)). The shade of the yellow highlighting on each term reflects $\phi_{t,j}$, the term's probability for the corresponding narrative.

(a) Table 4

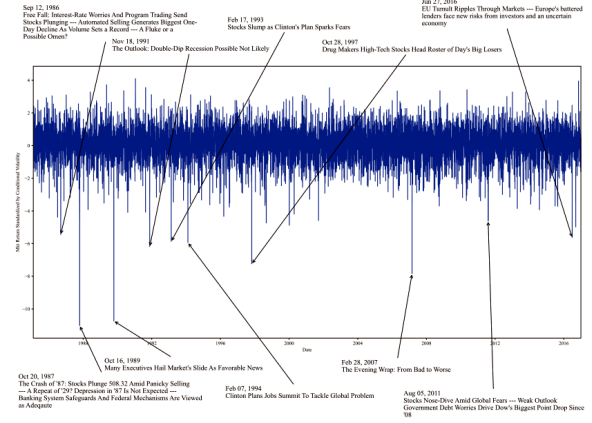


Figure 9
Event retrieval for market returns

Blue curve: Daily market returns adjusted by conditional volatility, $\frac{r_t^{\text{Mkt}}}{\sigma_t^{\text{Mkt}}}$, where σ_t^{Mkt} is EWMA volatility. We tag large spikes with retrieved news events. The article titles are reported in this figure, and the text bodies are in [Internet Appendix D.2](#).

(b) Figure 9

8 Result map

- Sparse IPCA selects a small subset of narratives and improves MVE Sharpe ratios.
- Narrative factors reduce pricing errors on anomaly portfolios and size/book-to-market portfolios.
- Narrative MVE portfolios outperform benchmark MVE portfolios out of sample.
- Narrative factors add incremental value to FFC6.
- The narrative pricing kernel predicts macro variables with ICAPM-consistent signs.
- Recession, Record High, and Optimism provide central economic interpretations.

9 Short summary

- WSJ narratives are converted into 180 topic attention series.
- Narrative attention shocks are linked to stock returns through covariance instruments.
- Sparse IPCA estimates priced latent factors and selects relevant narratives.

- Narrative factors perform well in cross-sectional pricing and out-of-sample MVE tests.
- The narrative pricing kernel forecasts macro variables in ICAPM-consistent directions.
- Model interpretation links latent factors back to topics, words, and articles.

10 Conclusion

10.1 Core conclusion

Business-news narratives can be transformed into interpretable systematic risk factors. The resulting factors price assets, form high-Sharpe MVE portfolios, and behave like ICAPM state variables.

10.2 Economic intuition

Investors use business news to update beliefs about wealth and investment opportunities. Assets that covary with these belief updates should earn risk premia.

10.3 Methodological contribution

The paper combines LDA, IPCA, and group lasso to construct a text-based, interpretable asset pricing model.

10.4 Limitations

The design does not prove news causality, topic labels require interpretation, and the results depend on the WSJ corpus and factor-model assumptions.

10.5 Takeaway

Text can be a disciplined source of interpretable risk factors that connects empirical asset pricing to ICAPM theory.